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Prediction of CHF location through applied machine learning

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ABSTRACT

The prediction of critical heat flux (CHF) location is a challenging issue in power generation industries. Several factors *viz.* geometric parameters and operating conditions influence the occurrence of CHF, and many mechanisms have been proposed to better explain the physics behind the occurrence of CHF. As CHF is a complex phenomenon, its dependency on these factors exhibit non-linear relationship. Further, it is challenging to account for all the factors simultaneously in a single model. Moreover, the benchmark experimental data has to be available to validate the constructed model with all the factors *viz.*, operating pressure and power, inlet mass flux and subcooling *etc*. Due to these reasons, the influence of these factors remain unresolved. Considering the above issues, the present study employs different machine learning models to predict CHF location and identify the accurate model based on the comparisons against the chosen benchmark experimental data. Becker et al. (1983) data bank was selected for machine learning model training as well as the testing due to the wide range of operating conditions covered. Hyperparameter tuning is applied to optimize the performance of the models. Among all the implemented models, the artificial neural network (ANN) model is found to perform well with the training and the test datasets. The mean absolute percentage error (MAPE) of the training and testing data points are 4.01% and 6.04%, respectively. The accuracy score of the ANN model is approximately 96% for the training and 94% for the testing datasets. The proposed model is helpful for design engineers and analysts of power plants for design optimization and to reduce the computational time.

1. Introduction

Boiling heat transfer has a wide range of applications from everyday life to industrial domains. The study of boiling heat transfer has been extended to the domain of micro scale heat transport (Cahill et al., 2003; Mukherjee et al., 2018). To meet the increasing power requirements, the coolant phase change is one of the most sought after alternatives for the electronic cooling applications as well as power plants to improve the thermal efficiency. Although the coolant phase change exploits the latent heat to enhance the heat transfer rates, the generated vapour phase hampers the heat transfer rate which is detrimental to the structural integrity of the thermal systems. Boiling crisis or critical heat flux (CHF) in a two phase flow boiling system is detrimental to the design of thermal components in many engineering systems, such as high-power microprocessor cooling, boilers, chemical plants, petroleum refineries, and most often nuclear power plants. Fig. 1 depicts the schematic of the two phase flow boiling regimes in a vertical circular tube, and the corresponding heated wall and bulk temperature variations along the axial length of the circular tube. The heat transfer coefficient at the heated tube wall and the coolant interface drops down sharply as the vapour phase completely covers the heated surface due to liquid film dryout. The corresponding operating power is called the CHF or dryout power. Therefore, the maximum operating power is limited below the CHF value to avoid catastrophic failure of the thermal systems. To this end, accurate design and modelling of thermal systems are necessary to ensure safety and operational efficiencies. Therefore, it is essential to understand the boiling heat transfer and accurately predict the CHF power or location in thermal systems that undergo coolant phase change to ensure safe working conditions within the design limits.

Many researchers and design engineers have extensively focused on the study of CHF to predict the safety limits. Performing experimental studies (Bottini et al., 2020) to understand the boiling phenomenon and to predict the CHF are challenging. While employing the intrusive techniques, the thermocouples, probes *etc*., act as the nucleation sites for the vapour generation. Whereas using the non-intrusive techniques,

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the generated vapour masks the field view of the camera, and obstructs the capturing of the phase change phenomenon in the vicinity of the heated wall. The theoretical or analytical solutions (Pothukuchi et al., 2021; Du et al., 2012) are possible with the help of highly simplified assumptions for the regular geometries such as, circular tubes, annulus etc. However, the solutions based on such assumptions are not suitable for the thermal systems with complex geometries. CHF look-up tables (LUTs) are developed based on the experiments performed in an 8 mm diameter circular tube (Groeneveld et al., 2007). For a given operating conditions, the CHF can be identified by an iterative procedure followed by interpolation methods. But, suitable correction factors have to be used depending on the complex geometries considered. The complex geometries of rod bundles with spacer grids, mixing vanes *etc*., can be easily handled by sub-channel analysis codes (Pothukuchi et al., 2016) which are generally one-dimensional in nature. Although three dimensional subchannel analysis codes are developed (Mohitpour et al., 2014; Salko et al., 2019), they are heavily dependent on empirical correlations. Further, the local velocity and temperature distributions cannot be obtained. The alternative is to perform the three dimensional computational fluid dynamics (CFD) simulations based on the mechanistic models to obtain the detailed local flow and temperature distribution (Pothukuchi et al., 2019; Gui et al., 2020; Song et al., 2020). The limitation is that the mechanistic models are not valid for all the flow boiling regimes, and the prediction accuracy depends on the choice of closure models (Wang et al., 2023). Moreover, there is no generalized and robust modelling framework that can identify different mechanisms of CHF and accurately predict its occurrence. Despite numerous experimental and modelling based studies, boiling heat transfer is a very complex phenomenon in which the occurrence of CHF is influenced by a variety of factors such as, operation conditions (Misale et al., 2009), thermal characteristics of fluids and boiling surface materials (Bar-Cohen et al., 2006), surface morphologies (Ferjančič and Golobič, 2002), geometrical configurations (Kandlikar, 2002; Liu et al., 2018), and the surface oxidation of boiling surfaces (Kang et al., 2018). Accounting for the effects of multiple factors on CHF location in single individual models is challenging. Therefore, Machine Learning (ML) algorithms are naturally preferred as a viable option for the prediction of CHF location.

To this end, the paper is organized as follows: Section 2 presents the literature review; Section 3 describes the dataset chosen; Section 4 outlines the data pre-processing and model performance evaluation basis; Section 5 covers different machine learning (ML) models that are implemented; Section 6 analyses these ML methods based on the comparison against the experimental data, and the concluding remarks are listed in Section 7.

2. Literature review

For the last 50 years, with the development of nuclear reactors, tremendous attempts have been invested in understanding the complexity of the CHF phenomena. The methods developed so far for the prediction of CHF values include empirical correlations, analytical models (Pothukuchi et al., 2021; Wang et al., 2023), look-up tables (Groeneveld et al., 2007), and detailed computational fluid dynamics simulations (Pothukuchi et al., 2019). Each method has its pros and cons owing to the highly complex and empirical relations, phenomenological complexity, and unpredictability of CHF occurrence. The empirical correlations are easy to implement but limited due to highly complex correlations and often fail beyond their ranges of application ability. The analytical CHF models rely on too many assumptions and are empirical. The CHF look-up table methods are based on the CHF data for an 8 mm diameter tube and require interpolation or extrapolation for the intermediate values in the table. The detailed computational fluid dynamics simulations are computationally extensive, and the prediction accuracy depends on the choice of closure models (Pothukuchi et al., 2019). The limitations of the conventional CHF predicting approaches are listed in Table 1. To overcome these limitations, models based on machine learning methodology provide an alternative approach to predict CHF. Using the experimental data or data generated by the subchannel analysis codes and selective CFD simulations, ML model that is valid for a wide range of operating conditions can be developed.

Table 1
Limitations of the conventional CHF prediction approaches.

CHF prediction methods	Limitations
Experimental studies	Expensive, time consuming and intrusive
Analytical models	Simplified assumptions and empirical
LUT	Based on 8 mm circular tube data
Subchannel analysis	Reduced order approach and empirical
CFD simulations	Time consuming and sensitive to the closure relations

Machine learning methodology, unlike conventional approaches based on physics, is a data-driven statistical method that solely responds to training datasets; the assumptions do not require a physics foundation, but they must be valid for the requirements of training datasets. In terms of feature distributions, training sets are meant to be unbiased, implying that the training data size for each feature should be uniformly distributed over the total training data. However, each modelling challenge has a prediction model built on a unique ML structure. For example, a multi-layer forward feed neural network provides the best estimation of thermal conductivity of nanofluids (Esfe et al., 2017). The heat transfer coefficient could be modelled for nanofluid pool boiling using an adaptive neuro-fuzzy inference system (Salehi et al., 2013). While the support vector machine (SVM) appears to be the ideal modelling framework for CHF modelling of nuclear fuel rods in light water reactors (LWRs) (Jiang and Zhao, 2013a,b).

In the literature, several ML based methods were proposed for the study of two phase flow systems focusing on boiling incipience, flow regime identification till the prediction of CHF. Table 2 presents a brief overview of two phase flow studies based on ML algorithms. Ueki and Ara (2021) have employed deep neural network (DNN) techniques to predict the boiling incipience based on the pressure acoustics. The type of flow pattern is critical for calculating system pressure accurately (Almalki and Ahmed, 2020). A probability density function of the time signal for void fraction and a high-speed imaging system was used to identify the downstream flow pattern. All studies were presented using the estimated superficial velocity of the mixture based on the area of the orifice utilized and the volumetric quality. The expected flow pattern downstream of the orifice was determined using the Classification Learner Environment (CLE) algorithm with a total error of 9%. Hernandez et al. (2023) have employed the conformal ANN techniques to enhance the prediction accuracy of the vapour volume fraction in a helical coil tube whereas Ooi et al. (2022) have identified different flow boiling regimes in a vertical annular channel using artificial neural networks (ANN) and K-nearest neighbours (KNN) assisted SVM ML algorithms. They have covered only the bubbly, transition (cap-bubbly), and slug flow regimes since reaching churn and annular flow regimes would necessitate high heat flux and low subcooling conditions, which could damage their experimental test facility. Their primary goal was to use self-organizing maps to identify local and global flow regimes in boiling flows using processed conductivity probe signals, with a secondary purpose is to extract features from the data that can be utilized to train ML techniques like KNN and SVM to classify flow regimes. ANNs correctly identify flow regimes having an accuracy ranging from 96% to 100%, but clustering techniques fared poorly with an average accuracy of only 53%.

Zhao et al. (2018) proposed a hybrid physics-informed ML aided framework to predict CHF corresponding to departure from nucleate boiling (DNB) type CHF database. They have implemented ANN and random forest (RF) models to predict the DNB type CHF in subcooled or low-quality flow boiling regime. The experimental CHF dataset was

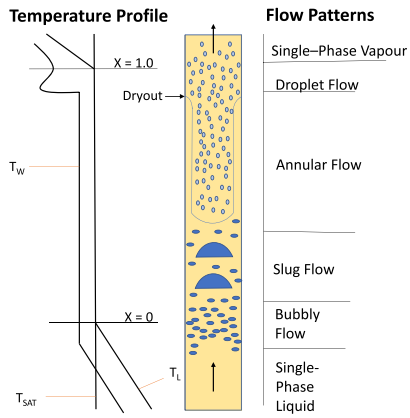

Fig. 1. Schematic of two-phase flow boiling in a vertical heated channel indicating different flow boiling regimes and the variation of wall and bulk temperatures.

Fig. 2. Heat map correlation for different parameters.

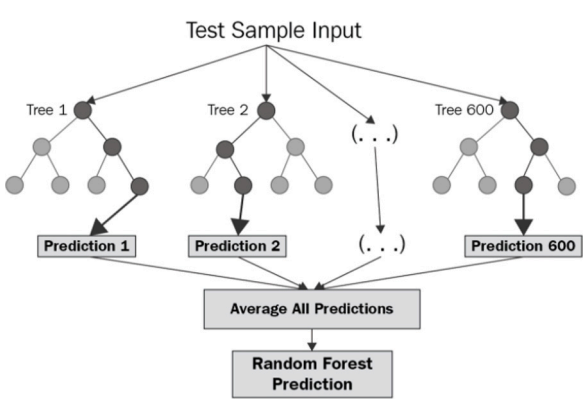

Fig. 3. Schematic of a Random Forest tree.

formed by collating 1865 test cases including round tube, annulus and heated plate on one side. Exit pressure, mass velocity, local (exit) equilibrium quality, equivalent and heated diameter and heated length are input features derived from the raw data, whereas CHF values are the target (outputs). Training errors (RMSE) are extremely low as 7.2% and 5.2% with ANN and RF respectively, while validation errors (RMSE) are around 14% with both ANN and RF (Zhao et al., 2020). He and Lee (2018) used the mass flux, equilibrium quality and pressure as their dominant parameters to predict CHF and reconstruct the CHF dataset using υ-SVM. In their further research, He and Lee (2020) showed that Deep Belief Network performed better predicting the CHF on microstructure surfaces. This work provides a unique method for predicting CHF on microstructure surfaces by employing DBN for CHF regression. The findings reveal that DBN is capable of resolving the complex parametric behaviour of CHF in terms of important design factors such as height, diameter, and spacing. The method is predicted to aid in microstructure design optimization for CHF maximization. Gaussian process regression (GPR) and ant colony optimization (ACO) techniques are also used to predict the CHF (Jiang et al., 2020). Due to flexibility, adaptability and powerful applicability in solving complex nonlinear situations, GPR has recently attracted more attention. GPR has been used to solve various problems such as, estimating safety margins in mechanical engineering (Fynan and Ahn, 2016), modelling nuclear-component degradation (Baraldi et al., 2015), online monitoring of sensors (Lee and Chai, 2019) and condition monitoring of a component (Vitelli and Zio, 2013). However, it is currently less well recognized in the field of CHF prediction, which can be explored in future research. ANN is applied to quantitatively predict the CHF for narrow rectangular channels in a steady state condition (Kim et al., 2021). Datasets of the rectangular channel are scarce to train the neural network. To address this issue, they created pseudo-data for neural network training by taking into account their relevant flow and dimensional conditions rather than experimental data. The proposed network was trained in two stages. The deep belief network (DBN) was first trained to identify data attributes. As DBN is an ML model with no convergence constraint, it was trained across 3000 epochs long enough to achieve convergence with a provisional divergence loss. Second, they used the DBN results and a batch size of 500 to train the ResNet to predict CHF. The trained ANN predicted the test pseudo-data with an error (RMSE) of 17.2%.

Table 2
Brief literature review for the ML based studies.

Authors	Methodology	Remarks
Ueki and Ara (2021)	DNN applied on the pressure acoustics data	Boiling incipience
Almalki and Ahmed (2020)	Reconstructing CHF dataset by υ-SVM and predicting CHF	Recognition of two-phase flow pattern
Ooi et al. (2022)	KNN and SVM	Flow regime identification
Hernandez et al. (2023)	ANN with CLOGSIG	Volume fraction in a helical coil tube
Zhao et al. (2018)	DFF, NN and RF	CHF prediction
Zhao et al. (2020)	Physics Informed ML	CHF prediction
He and Lee (2018)	Reconstructing CHF dataset by υ-SVM	CHF parametric study
He and Lee (2020)	Deep Belief Network	CHF prediction on micro-surfaces
Jiang et al. (2020)	GPR-ACO and SVR	CHF prediction
Kim et al. (2021)	Neural networks	CHF prediction
Park et al. (2020)	Big Data of CHF by SPACECODE and NN	Wall temperature prediction at CHF

Table 3
Parameters and their range.

Parameters	Range
Mass flux (G, kg m^{-2} s^{-1})	496–3493
Heat flux (q_w, MW m^{-2})	0.147–60.5
Pressure (P, MPa)	2.98–119.8
Tube inner diameter (d_i, m)	0.01–0.0247
Subcooling (ΔT_{sub}, K)	4.6–20.7
Heated tube length (L, m)	7–7.1
CHF location (z_{cr}, m)	0.63–6.75
Critical quality (x, –)	0.03–1.66

Park et al. (2020) predicted the wall temperature corresponding to a CHF location using artificial neural networks (ANN), enhancing the process calculation speed. The SPACE algorithm determines the heat transfer regimes of the pre- and post-CHF by calculating the wall temperature from a nucleate boiling heat transfer model at the given CHF. It requires iterations and a significant amount of computing time. This work introduces the use of a machine learning algorithm to reduce calculation time. The subprogram developed in the SPACE code was used to create big data on the wall temperature at CHF. The ANN using the SPACE code correctly predicted the wall temperature at CHF. The computation time was also investigated during the test calculations. All the suggested ANN models significantly improved calculation speed, resulting in a time savings of up to 86%.

From the literature review, it is evident that there is no generalized method that is robust and applicable to a wide range of operating conditions for predicting CHF or its location. Therefore, it is essential to devise an efficient method that overcomes the limitations of the conventional approaches (see Table 1), and obtain accurate predictions of CHF location. With the advancements in computational capabilities and optimization techniques, ML have also received fast-growing interest in the field of thermal-fluid systems. It is observed in the literature that ML is a widely employed tool for building complex models with many different dependencies for predicting the CHF and its location. However, the data samples used for constructing ML models are either small or based on LUTs (Groeneveld et al., 2007). In the present study, experimental data of Becker et al. (1983) which covers a wide range of operating conditions is chosen for the development of ML model to accurately predict the CHF. To the best of authors' knowledge, ML algorithms are not applied on this experimental CHF dataset. Therefore, the present study aims at the development and assessment of different ML models for the prediction of CHF location based on the experimental CHF dataset of Becker et al. (1983).

3. Dataset description

Identification of datasets is very important for the development of a ML model for the prediction of CHF. Various parameters such as operating conditions, thermal characteristics of working fluids and boiling surface materials, surface morphologies, geometrical configurations *etc.*, are commonly acknowledged as determining factors for CHF and its location. Although multiple variables influence the occurrence of CHF, it is very challenging to conduct experiments that cover a wide range of operating conditions. Furthermore, there are not many datasets available in the literature other than the data generated by LUTs. To this end, in the present study, experimental data of Becker et al. (1983) is chosen. Becker et al. (1983) have conducted experiments to predict CHF in a circular tube. This experimental CHF dataset has a total of 421 data points which covers a wide range of operating conditions *viz.*, mass velocity or mass flux, pressure, heat flux, the inner diameter of the tube, heated length, inlet subcooling and CHF location. Table 3 presents the operating parameters and their ranges in Becker et al. (1983) dataset. The present work considers six input parameters – operating pressure, wall heat flux, inlet mass velocity, inlet subcooling, inner tube diameter and heated length (Becker et al., 1983), whereas CHF location is the target feature. The chosen benchmark dataset is divided into two parts: training and testing sets. The training set includes 80% of the samples to train the machine learning model, whereas the testing set includes 20% of the samples to test the performance of the model.

3.1. Effect of different factors on CHF location

One of the most heavily debated aspects of boiling heat transfer is the prediction of CHF and its location over a heated surface. CHF occurs when a vapour layer covers the heat transfer surface and marks the upper limit of efficient boiling. Multiple parameters, including the thermal properties of coolant, heat transfer substrates, interfacial properties, and system conditions, have been found to influence the CHF and its location. In the present study, all the available parameters *viz.*, heated length, diameter, operating pressure, wall heat flux, inlet mass flux and inlet subcooling are used for the prediction of CHF location in a vertically heated circular tube.

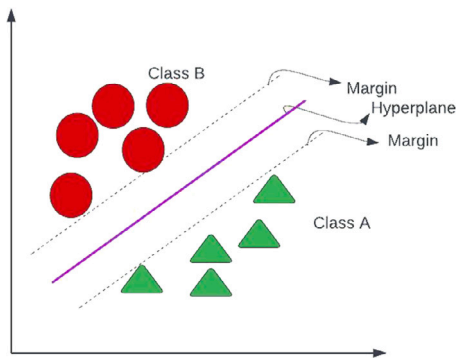

Fig. 4. Schematic of Support Vector Machine hyperplane with margins.

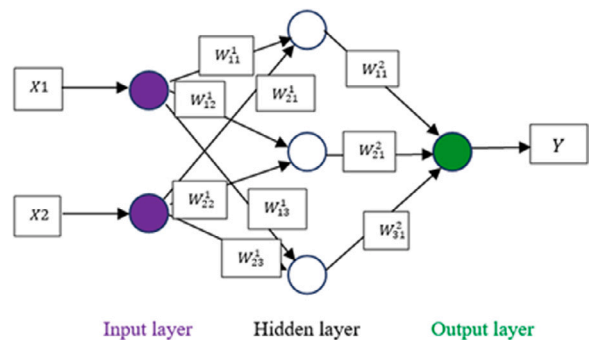

Fig. 5. Schematic of general ANN architecture.

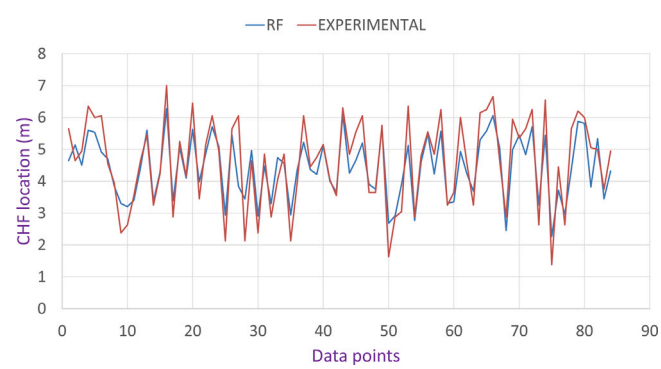

Fig. 6. Comparison of experimental and predicted CHF location for RF model.

3.2. Heat map correlation

The Pearson correlation coefficient is the measure of the linear relationship between two features. Given paired data $(x_1, y_1), \ldots, (x_n, y_n)$ consisting of n pairs, the mathematical expression of calculating the Pearson coefficient is as follows:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{1}$$

where, r is the correlation coefficient, n is sample size and x_i and y_i are values of x-variable and y-variable in the ith indexed sample

respectively. $\bar{x}$ and $\bar{y}$ are the mean of the values of x-variable and y-variable respectively.

The correlation heat map for the Becker et al. (1983) dataset with Pearson correlation coefficient values is presented in Fig. 2. It is the product of standard deviations of the two features and then divided by their covariance. The value of the Pearson correlation coefficient ranges from −1 to +1. A larger absolute value of the Pearson correlation coefficient indicates stronger linear relation, and a smaller absolute value of it indicates a weaker linear correlation.

Positive and negative correlation implies that one variable increases as the other increases, and one variable decreases as the other increases. The zero value of the Pearson correlation coefficient suggests that there is no linear relationship. According to the correlation heat map in Fig. 2, length, diameter, pressure and heat flux are moderately influential factors in predicting CHF location. In contrast, mass flux and sub-cooling temperature difference have a negligible linear correlation with CHF location. Length and diameter are the independent features which are highly correlated. In this problem, the number of input features available in the given dataset is very small, therefore we include all of these features for training the machine learning models.

4. Data pre-processing

In the present study, ML models are implemented to predict the CHF location using Becker et al. (1983) dataset comprising of six input parameters and one target feature. Input features include operating pressure, mass velocity, heat flux, inner diameter, heated length, and inlet subcooling. The target is the values of CHF location (output). Becker et al. (1983) is a dataset in raw form having null values and outliers which restrict the learning of ML models to learn properly and that result in overfitting or underfitting of the models. The dataset is having a total of 412 data points.

It has been shown in studies that data pre-processing techniques like normalization and standardization aid in improvising the learning mechanism of machine learning models. These models perform better and converge faster when datasets are scaled, as doing so avoids a bias towards a particular feature. For this study, scaling was done using Standard Scaler on the Becker et al. (1983) dataset. Standard scalar standardizes a feature by subtracting its mean value from each data point and then, dividing it by the standard deviation. It results in a distribution where the Standard Deviation of each feature is 1, and the mean is approximately 0.

$$X' = \frac{(X - \mu)}{\sigma} \tag{2}$$

where μ and σ represent the mean and standard deviation of the feature values respectively. X and X' are the original and transformed values respectively.

4.1. Feature selection

Feature selection is a crucial step in machine learning since it eliminates redundant and irrelevant data. It results in reduction in training time, memory consumption and better performance. There are many feature selection techniques discussed in the literature, but in the present study, minimum redundancy maximum relevance (mRMR) is chosen (Hanchuan Peng and Ding, 2005).

The mRMR algorithm ranks features according to their maximum correlation with the output variable and with minimum correlation between other input features. F-statistic metric is used by mRMR to determine how dependent a feature is on the output variable (relevance), while pairwise correlation metric is used to determine how closely a feature is related to the other input features (redundancy). mRMR selects one feature at a time while working iteratively. The feature with the highest score among the remaining unselected features is determined at each step using either a difference (relevance minus redundancy) or a quotient (relevance divided by redundancy) approach.

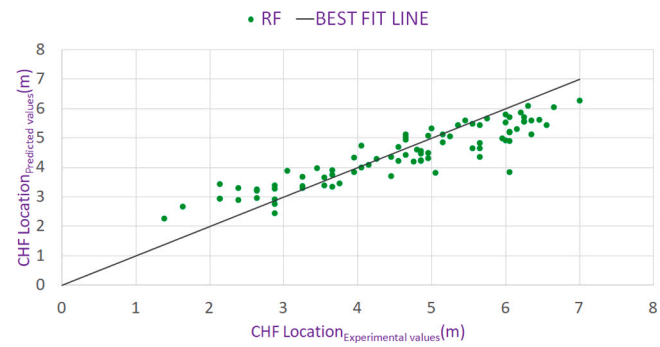

Fig. 7. Scatter plot of RF model with best fit line.

mRMR score for all the independent features is tabulated in Table 4 which shows that mass flux is highly correlated with the output variable (*i.e.*, CHF location) and least with any other independent feature. In contrast, tube length and tube inner diameter have low mRMR scores. The experimental results mentions model results evaluated after removing these features.

4.2. Model performance evaluation

ML algorithm constructs probabilistic models to understand patterns from the training data, and then the model is used to predict the test (unseen) data. Fundamentally the goal is to create quality models with good predictive power. The mean square error (MSE) is one of the performance criteria used to evaluate the predicted accuracy along with mean absolute percentage error (MAPE), and coefficient of determination (R^2) (Ali et al., 2012). These indices are used to see how closely the predicted values match the experimental values. MSE (Tang et al., 2017) is the average of the squares of the difference of experimental values and predicted values. The mathematical expressions are as follows:

$$MSE = \frac{1}{N} \Sigma_{i=1}^{N} (y_i - \hat{y}_i)^2 \tag{3}$$

where N is the total number of data points, $\hat{y}_i$ is the predicted and y_i is the experimental CHF location (see Fig. 14).

MAPE is also used to measure the error of the model in terms of loss function of experimental and predicted values. The mathematical expression of the MAPE is given below.

$$MAPE = \frac{1}{N} \Sigma_{i=1}^{N} \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100 \tag{4}$$

where N is the total number of data points, $\hat{y}_i$ is the predicted and y_i is the experimental CHF location.

R^2 score is the statistical measure of the goodness of fit for machine learning models and that is calculated as below equation.

$$R^2 = 1 - \frac{\Sigma_{i=1}^{N} (y_i - \hat{y}_i)^2}{\Sigma_{i=1}^{N} (y_i - \bar{y})^2} \tag{5}$$

where $\bar{y}$ is the mean of the experimental CHF location values. Model performance is evaluated in Section 6.

5. Machine learning models

Based on the literature review, three different ML models *viz.* Random Forest, Support Vector Machine and Artificial Neural Network have shown convincing results in the context of two phase flows. Therefore, in the present study, these three ML models are chosen to predict the CHF location.

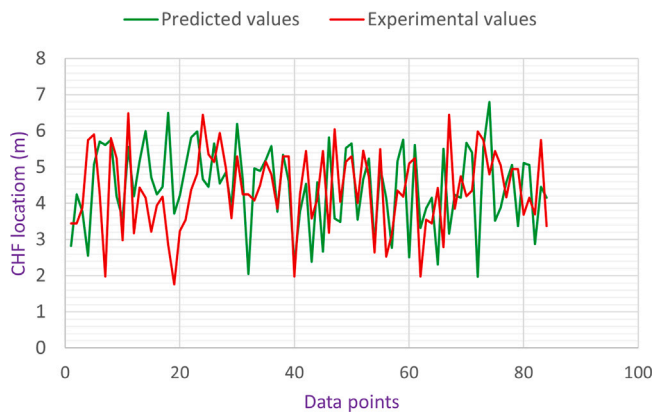

Fig. 8. Comparison of experimental and predicted CHF location for SVM model.

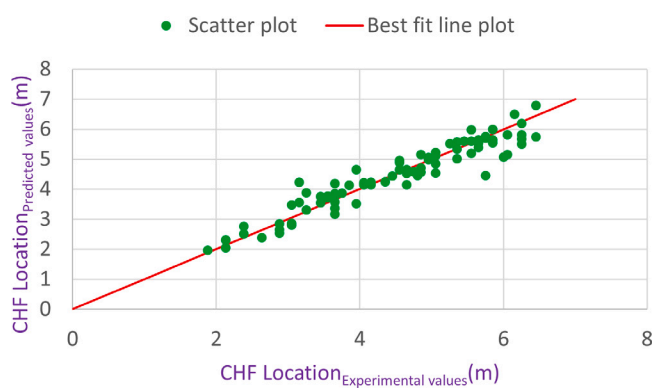

Fig. 9. Scatter plot of SVM model with best fit line.

5.1. Random Forest (RF)

The Random Forest (RF) algorithm works by building multiple decision trees and uses the output of each individual tree to give a cumulative result. Decision trees are built by splitting the datasets into two parts at each node. Metrics like Gini gain or information gain are used to decide which feature to use to split the dataset.

$$G = \sum_{i=1}^{N} P_i \cdot (1 - P_i) \tag{6}$$

where G is the Gini impurity and P_i is the probability of a particular feature having a class i. The more the gain, the better the feature is for splitting the data. The problem with simple decision tree is that they are prone to overfitting. However, in random forest, many decision trees are created that reduce the variance of the output.

In order to create a tree in random forest, a fixed number of data points are chosen randomly from the dataset. To increase the randomness further, a random set of features are tested with the gain metric. Doing so avoids including features that have a very high predictive power in every tree, which in turn creates many uncorrelated trees. This process is done multiple times to create a huge forest of trees. Therefore, a diverse set of trees are made that have low variance compared to a single decision tree. The architecture of random forest model is shown in Fig. 3. The output of the random forest for the regression problem is the average of the output of every single decision tree, which is known as bagging. As a result, random forest has the advantages *viz.*, solving the challenge of overfitting, less sensitive to outlier data in the training dataset, parameters can be simply set, obviating the need for tree trimming, variable importance and accuracy are automatically generated *etc*. RF not only retains the benefits of decision trees, but it also shows superior results than decision trees due to bagging and voting mechanism (Qi, 2012) by which decisions are determined, and a random selection of variables.

5.2. Support Vector Machine (SVM)

SVM is a parametric machine learning model that can solve linear and non-linear classification and regression problems. The SVM algorithm aims to find a hyperplane in an N-dimensional space that classifies the data distinctly. The hyperplane is in (N-1) dimensional subspace corresponding to a N-dimensional feature space. For a 2-D feature space, *i.e.*, with only two input features, a hyperplane corresponds to a line, while for a 3-D space, it can be a 2-D plane.

Consider an equation of a hyperplane in case of Fig. 4, a line:

$$w^T \phi(x) + b = 0 \tag{7}$$

where $\phi(x)$ is the input feature matrix, w (parameter terms) and b (intercept term) are coefficients of the hyperplane equation. Consider a binary classification problem for two feature input, which has two classes labelled as $y = -1$ and $y = 1$. For correct classification in a perfectly linearly separable dataset:

$$y_n \left[w^T \phi(x) + b \right] \geq 0 \tag{8}$$

where y_n is class label of every nth data point.

Thus in a case, where the data is linearly separable, a hyperplane is identified by the model that best separates the two classes of data. Apart from ensuring that every observation is on the correct side of the hyperplane, SVM also tries to maximize the perpendicular distance of this hyperplane from the nearest observations (called support vectors), called the margin.

The SVM optimization problem is a constraint optimization problem. The Lagrange multiplier method is employed to solve this constraint optimization problem. Since the dataset cannot always be perfectly linearly separable, SVM uses a slack variable (similar to regularization) to penalize every miss-classification. Apart from the slack variable (also called soft margin) to address the non-linearity of the dataset, kernel trick (utilizing existing features, applying some transformations, and creating new features) is employed to help SVM find the non-linear decision boundary. Multiple hyperparameters (like the coefficient of slack variable (C), kernel type, *etc*.) need to be tuned while fitting an SVM model to a particular dataset. A distinction between random forests and SVM models is that random forest is a non-parametric model, whereas SVMs are parametric in nature. Thus, it is able to identify the parameters that best suit a given dataset.

5.3. Artificial Neural Network (ANN)

Artificial Neural Network is another category of parametric machine learning models which are useful for modelling nonlinear and complex relationships in data. Among the different types of ANNs, the feed-forward neural networks (FFNNs) are found to be robust in learning complex relations in data and widely used in different engineering applications (Kim et al., 2021) such as resource usage prediction in cloud data centres, speech processing *etc*. FFNNs follow a layered architecture with three main layers: input layer, hidden layer(s) and output layer. Input layer accepts the input information from the dataset and passes it to the next layer called a hidden layer(s). The number of neurons in the input layer is equivalent to the number of features in the dataset. The hidden layer(s) can be either one or more in number depending upon the complexity of the data points. They pass their outputs to the final layer that give the final output of the model. All the layers consist of neurons that are interlinked through connections having weights and biases. It receives input from the preceding layer neurons, processes it, and produces an output.

Activation Functions are used at each node to introduce non-linearity. Without them, the data would pass through the nodes of the

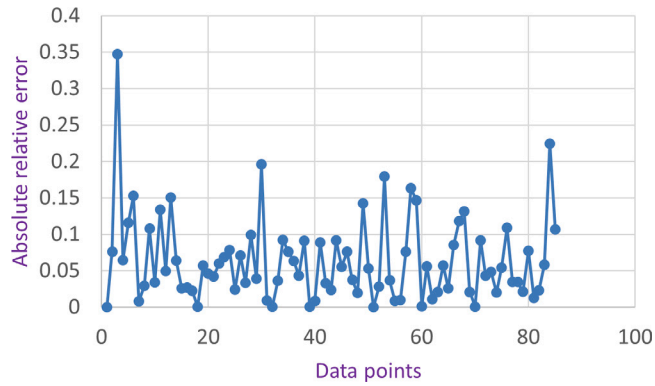

Fig. 10. Absolute relative error for different datapoints obtained using SVM model.

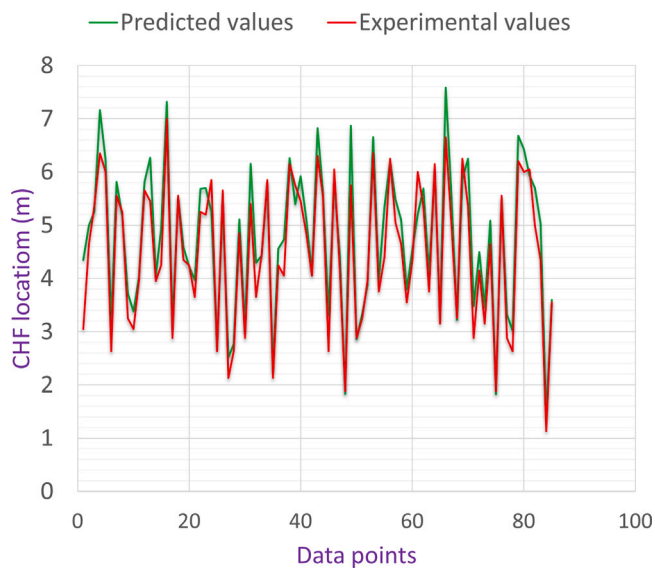

Fig. 11. Comparison of experimental and predicted CHF location for ANN model.

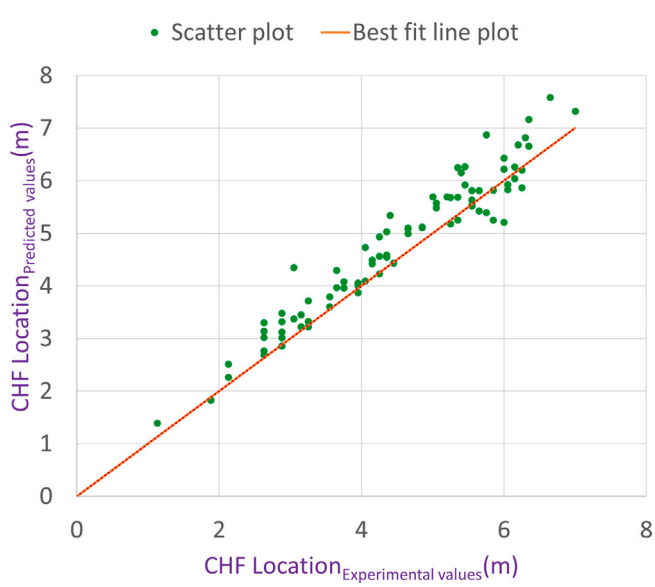

Fig. 12. Scatter plot of ANN model with best fit line.

Table 4
mRMR score of all independent features.

Parameters	mRMR score
Mass flux (G, kg m^{-2} s^{-1})	248 633.778
Heat flux (q_w, MW m^{-2})	2.146
Pressure (P, MPa)	3.582
Tube inner diameter (d_i, m)	0.034
Subcooling (ΔT_{sub}, K)	4.822
Heated tube length (L, m)	0.560

Table 5
Quantitative performance measure of RF.

Stages	MSE	MAPE	R^2
Training	0.0689	0.0540	0.9530
Testing	0.4233	0.1299	0.7761

Table 6
Quantitative performance measure of SVR with mRMR feature selection.

Number of mRMR features used	MSE	MAPE	R^2
6	0.169	0.448	0.870
5	0.373	0.852	0.713
4	0.451	0.810	0.654
3	0.879	1.566	0.326

network only going through linear functions and the algorithm will be unable to build complex models. Eq. (9) shows the output of a layer in a neural network and Eq. (10) shows the Rectified Linear Unit (ReLU) activation function.

$$Y = f(B + \Sigma_{i=1}^{n} W_i \times X_i) \tag{9}$$

$$f(z) = max(0, z) \tag{10}$$

where, B is the bias, W is the weight and X is the input feature and Y is the output.

The schematic of fully connected ANN architecture is presented in Fig. 5. In the figure, $W_{x,y}^{z}$ represents weight of the link between node x in layer z and node y of layer $z+1$. For example, $W_{3,1}^{2}$ refers to the weight of the link between third node of second layer and first node of the third layer.

ANNs are trained through a continuous loop of forward and backward propagation. Initially, the values of the weights are randomly initialized. Then, the training data is batch-wise passed into the model, and the weight (or parameters) of the FFNN are updated in a single atomic step of forward and backward propagation. During the back propagation phase, the difference between the output value generated by the FFNN and the actual value is calculated using any loss function and the weights are adjusted by optimizing algorithms, like gradient descent, that aim to minimize the loss. The weights of the neurons in the last layer are adjusted first; therefore, this process is known as backward propagation. Once all the data points are presented to the model, one training epoch is said to be completed, and the whole process starts again (except random initialization). For this study, the mean squared error (MSE) was used as the loss function which is shown in Eq. (3). RMSProp (RMSProp, 2022) was used as the optimization algorithm for this application because of its quicker optimizing ability. The weights are adjusted according to Eqs. (11) and (12) in RMSProp algorithm.

$$v_{dw} = \beta \cdot v_{dw} + (1-\beta) \cdot dw^2 \tag{11}$$

$$W_{new} = W_{old} - \alpha \cdot \frac{dw}{\sqrt{v_{dw} + \epsilon}} \tag{12}$$

where v_{dw} is the moving average of squared gradients, α is the learning rate, β is a constant called momentum and dw is the gradient of the loss function with respect to W.

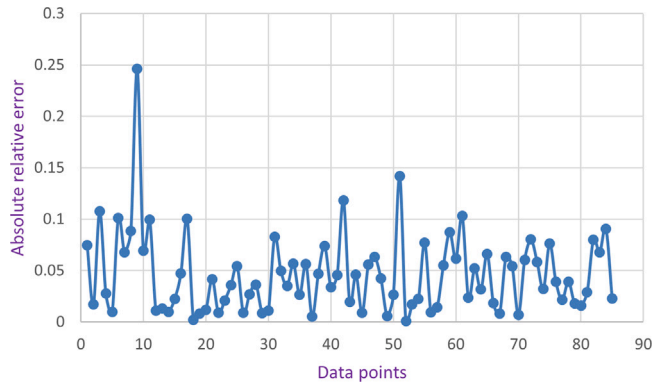

Fig. 13. Absolute relative error for different datapoints obtained using ANN model.

6. ML model predictions

The present study focuses on the development of machine learning model for the prediction of CHF location. This section discusses the prediction capability of RF, SVM and ANN models when compared against the experimental data of Becker et al. (1983).

6.1. Random forest

The RF model is implemented based on the experimental data of Becker et al. (1983). The RF is trained using Keras application programming interface (API) with TensorFlow (Antonio et al., 2019) backend and scikit-learn library in python 3.6. The number of training and testing data points are 337 and 84, respectively. The model framework is first trained from training dataset and then the performance is validated with the testing dataset. The number of trees is taken as 120. The performance of the model is checked by evaluating the MSE, MAPE, and R^2 scores. The quantitative performance of the model is tabulated in Table 5. The MSE for training is 0.0689 and for testing data sets is 0.4233. MAPE of train and test data sets is 0.0540 and 0.1299, respectively. The coefficient of determination (R^2) of the model for train and test data sets are 0.9530 and 0.7761 respectively. Fig. 6 shows the comparison of the predicted results against the experimental values. It is seen that for different data points, the experimental values and the RF model predictions are in good agreement. Fig. 7 presents the scatter plot of CHF location for experimental and predicted values of different datapoints. It is clear from figure that there is under predictions at some locations while over prediction at others. This occurs because the model is designed to approximate the overall trend of the data points, maintaining a consistent slope between the data points and the fitted line.

6.2. Support vector machine

As previously stated, numerous parameters need to be set for the prediction with SVR. Different combinations were tested before settling on a RBF kernel for the performance comparison with C = 230. However, in many circumstances, the polynomial kernel performs as well as an RBF kernel in our experience. SVR experiments were run in python 3.6. The SVM model was fit with the same hyperparameters after removing redundant features from the training and testing datasets using the mRMR feature selection technique. The results are tabulated in the Table 6. Fig. 8 shows the plot of the predicted and experimental values for different data points. Table 7 tabulates the quantitative performance of the model. The MSE and MAPE of the model for the training data points are below the 5% and for testing data points are around 13%. The R^2 score of the model is also above 90% for training and testing datasets. Fig. 9 shows the scatter plot of CHF location for predicted and experimental values. It can be noticed that the data points are very close to the best fit line. The absolute relative error of all the test data points is within 0.35 as shown in Fig. 10 and the average absolute relative error across all data points is 0.064.

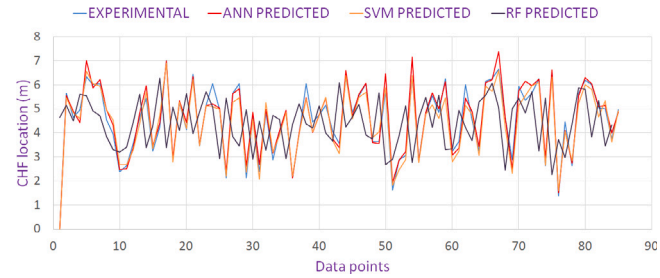

Fig. 14. Comparison of experimental and predicted CHF location using RF, SVM and ANN models.

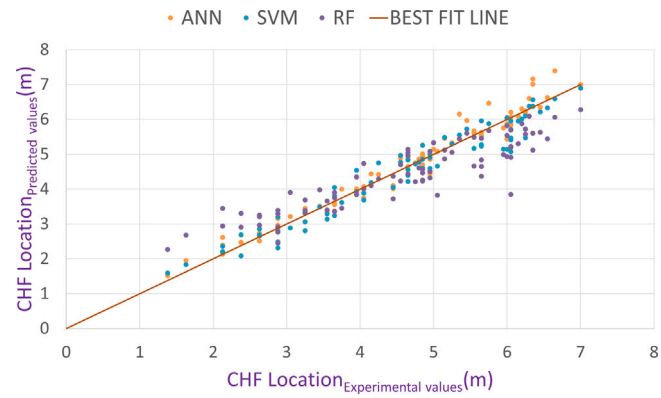

Fig. 15. Scatter plot of RF, SVM and ANN models with best fit line.

Table 7
Quantitative performance measure of SVR.

Stages	MSE	MAPE	R^2
Training	0.0482	0.0386	0.9697
Testing	0.1364	0.0644	0.9052

Table 8
Quantitative performance measure of ANN with mRMR feature selection.

Number of mRMR features used	MSE	MAPE	R^2
6	0.158	0.792	0.878
5	0.374	1.291	0.713
4	0.594	1.253	0.545
3	1.006	1.710	0.229

Table 9
Quantitative performance measure of ANN.

Stages	MSE	MAPE	R^2
Training	0.0474	0.0401	0.9676
Testing	0.1125	0.0604	0.9404

6.3. Artificial neural network

The ANN model is implemented based on the experimental data of Becker et al. (1983). The ANN is trained using Keras API with TensorFlow (Antonio et al., 2019) backend and scikit-learn library in python 3.6. The model framework is first trained and then the performance is validated with the testing dataset. It is seen empirically that the 6/50/50/50/1 architecture performs well for the data-set. Number of hidden-layers and neurons in those layers is decided experimentally.

Table 10
Quantitative performance analysis of the models (RF, SVM and ANN).

Stages	MSE	MAPE	R^2
Training	0.0689	0.0540	0.9530
Testing	0.4233	0.1299	0.7761
Stages	MSE	MAPE	R^2
Training	0.0482	0.0386	0.9697
Testing	0.1364	0.0644	0.9052
Stages	MSE	MAPE	R^2
Training	0.0474	0.0401	0.9676
Testing	0.1125	0.0604	0.9404

The ANN architecture 6/50/50/50/1 consists of 6 neurons in the input layer, three hidden layers with 50 neurons, and 1 neuron in the output layer. The tuned optimization algorithm is the RMSProp. The weights and biases are randomly initiated to their default values of random numbers and 0 respectively, and their best values are achieved by using the backpropagation method. Rectified linear unit (ReLU) is applied as an activation function on hidden layers. The number of epochs is taken as 1000 and batch size as 100. The ANN model was fit with the same hyperparameters after removing redundant features from the training and testing datasets using the mRMR feature selection technique. The results are tabulated in Table 8.

The quantitative performance of the model is tabulated in Table 9. The performance of the model is checked by evaluating the MSE, MAPE, and R^2 scores. The MSE for training is 0.0474 and for testing data sets is 0.01125. MAPE of train and test data sets is 0.0401 and 0.0604 which shows the ANN model is performing well for both labelled and unlabelled data points. R^2 score of the model for train and test data sets are 0.9676 and 0.9404 which indicates that the model is not over/under fitted. Fig. 11 compares the predicted results of ANN model against the experimental values. It is observed that for different data points, the experimental data and the ANN model predictions are in strong agreement. Fig. 12 presents the scatter plot of CHF location for experimental and predicted values. It can be noticed that the data points are near to the best fit line. The absolute relative error of all the test data points is within 0.25 as shown in Fig. 13.

6.4. Comparison of implemented ML models

The comparison of the predicted results of RF, SVM and ANN models against the experimental values is shown in Fig. 14. Fig. 15 depicts the scatter plot of CHF location for experimental and predicted values of the three models *viz.*, RF, SVM and ANN. The predicted data points are reasonably near the best-fit line. MSE and MAPE for all the explored ML models are tabulated in Table 10. The ANN model has the lowest training and testing error, and the evaluation scores of the ANN model are also better than the other ML models. The relatively lower performance of the Random forest regressor may be attributed to its inability to handle covariate shift (as it fails to extrapolate). Although the test scores of SVM and ANN are close, ANN performs better with a small margin. This further highlights the ability of ANN to capture the highly complex correlations within the CHF dataset.

7. Conclusions

The present study aims at employing different ML models to predict the CHF location covering a wide range of operating conditions. The CHF prediction time is much lower than the conventional CFD approach. The accurate CHF location predictions by the developed ML models would result in optimal design and analysis of thermal systems which prevent their catastrophic failure. The present study can be summarized into the following points.

- From the literature review, Becker et al. (1983) is chosen as the dataset for the ML model development due to the wide range of operating conditions covered.
- Different ML models *viz.*, RF, SVM and ANN models were implemented which have six input features and CHF location as the output feature.
- It is noticed that the ANN model performed well, and the MAPE of training and testing sets are found to be 4.01% and 6.04% respectively which conforms a close agreement to the experimental data.
- After models are fitted and redundant features have been removed, it was seen that the mRMR feature selection strategy did not produce superior outcomes. In both the ANN and SVM models, eliminating any feature made every evaluation metric worse. This trend can be attributed to the strong dependency of the output feature on the input features in the dataset. Every feature removal could result in significant information loss and degrade the performance of models. Although, the experimental dataset of Becker et al. (1983) covers a wide range of operating conditions, the size of the dataset is still considered smaller. The prediction accuracy of the CHF for a real power plant can be improved if a large dataset with different geometric and operating conditions are considered as the input features. This marks the need for the generation of CHF data based on the detailed CFD simulations after extensive validation studies.

CRediT authorship contribution statement

Vishnu Kumar: Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation. **Dhiraj Pimparkar:** Visualization, Investigation. **Vansh Rai Saini:** Visualization, Investigation. **Rishika Kohli:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Shaifu Gupta:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Harish Pothukuchi:** Writing – review & editing, Supervision, Resources, Formal analysis, Conceptualization.

Declaration of competing interest

None.

Data availability

Data will be made available on request.

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